

WHYALLA, Australia

WMO: 946640

Lat: 33.017S

Lon: 137.517E

Elev: 49

StdP: 14.67

Time Zone: 9.50 (AUA)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.5	41.6	24.8	18.9	71.1	28.5	22.5	70.7	23.1	62.0	21.2	60.8	3.4	320	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.1	99.8	68.0	95.3	67.2	91.1	66.3	73.4	87.7	71.7	86.6	70.1	83.8	12.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	109.8	78.5	67.3	100.5	77.2	65.6	94.5	76.1	37.0	87.8	35.5	85.9	34.1	83.9	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	20.3	18.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	74.5	76.5	70.6	64.9	58.6	54.9	53.9	55.3	59.1	63.6	68.0	70.8
	DBStd	9.66	7.62	7.62	6.29	5.48	4.45	4.29	4.07	4.79	6.17	6.64	7.45	6.15
	HDD50	20	0	0	0	0	0	5	8	6	1	0	0	0
	HDD65	1618	3	3	10	66	208	306	345	305	203	111	48	10
	CDD50	5181	760	742	638	448	266	151	129	168	273	421	540	645
	CDD65	1304	299	324	183	64	9	2	1	3	25	67	137	190
	CDH74	12319	2926	3210	1384	559	43	5	3	46	237	810	1431	1665
	CDH80	5675	1530	1587	572	180	4	0	0	5	61	333	663	740
(14) Wind	WSAvg	9.1	11.0	9.6	9.2	8.3	7.3	7.4	8.2	8.1	8.7	9.7	10.3	11.0
(15) Precipitation	PrecAvg	11.4	0.9	0.9	0.7	0.9	1.1	1.1	0.9	0.9	1.1	1.2	1.0	0.7
	PrecMax	21.1	4.5	5.6	4.4	4.1	3.7	3.6	2.2	2.2	4.3	4.1	3.4	3.6
	PrecMin	5.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0
	PrecStd	3.8	1.0	1.3	0.9	0.8	0.7	0.9	0.6	0.6	0.9	1.0	0.8	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.6	105.3	97.5	91.4	80.0	73.9	71.9	79.3	86.4	94.8	100.4	98.5
		MCWB	71.6	70.0	68.9	64.8	59.4	57.3	55.1	58.4	61.1	62.6	64.7	62.9
	2%	DB	99.5	100.2	90.8	85.0	74.3	68.4	67.7	72.9	80.4	88.2	92.8	93.2
		MCWB	68.9	68.9	67.1	61.8	57.7	57.0	53.4	55.6	59.4	59.8	63.4	65.0
	5%	DB	93.9	94.5	85.8	80.1	70.5	65.4	64.5	68.4	75.6	83.0	87.2	88.2
		MCWB	68.6	68.4	65.9	60.5	57.2	54.8	53.0	53.9	57.6	58.8	62.7	64.9
	10%	DB	88.3	89.4	81.2	75.4	67.2	62.9	61.6	64.7	70.8	77.3	82.4	83.6
		MCWB	67.7	68.3	65.0	59.9	56.7	53.7	52.0	52.9	56.3	58.4	62.3	64.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.9	76.4	73.4	68.3	64.2	62.8	58.6	61.1	66.6	66.5	69.6	72.3
		MCDB	89.3	84.7	91.8	77.5	69.0	65.5	66.5	72.7	79.8	79.8	88.0	84.8
	2%	WB	72.5	74.5	70.6	66.4	61.8	58.8	56.1	58.1	62.3	63.6	67.5	69.3
		MCDB	87.4	84.4	82.4	75.8	68.4	66.1	62.8	68.3	72.4	75.3	83.8	82.2
	5%	WB	70.9	72.5	68.7	63.9	60.0	56.7	54.3	55.7	60.0	61.7	65.7	67.6
		MCDB	86.0	87.1	78.5	75.1	67.5	63.7	61.4	64.4	69.8	73.6	79.4	80.0
	10%	WB	69.1	71.0	67.1	62.1	58.4	54.9	52.8	53.8	57.7	60.1	64.0	66.0
		MCDB	83.6	85.7	76.9	72.2	65.3	60.7	60.0	62.0	66.9	72.6	76.4	77.5
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	19.1	17.3	18.7	16.7	15.1	16.0	18.2	18.6	19.3	18.9	17.5
		MCDBR	30.2	28.0	25.7	27.1	21.5	18.6	20.5	25.0	27.7	31.9	30.0	27.6
		MCWBR	8.5	7.5	8.4	10.7	9.8	9.6	10.6	11.4	10.9	10.8	8.0	8.0
	5% WB	MCDBR	22.8	20.7	19.6	21.2	16.4	15.0	16.9	20.7	21.3	23.9	22.8	19.5
		MCWBR	8.4	7.4	8.8	10.9	9.0	9.8	10.5	11.2	10.3	10.6	7.8	8.7
(40) Clear-Sky Solar Irradiance	taub	0.348	0.347	0.328	0.320	0.308	0.297	0.297	0.308	0.335	0.343	0.349	0.340	0.340
	taud	2.537	2.562	2.602	2.605	2.615	2.624	2.618	2.573	2.491	2.476	2.498	2.543	2.543
	Ebn at Noon	314	308	303	289	276	272	277	288	296	305	312	318	318
	Edh at Noon	35	33	30	26	23	22	23	27	32	36	36	35	35
(44) All-Sky Solar Radiation	RadAvg	2546	2278	1891	1421	1024	852	928	1234	1674	2079	2360	2527	2527
	RadStd	124	118	99	67	41	47	42	68	108	104	105	111	111

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page